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Engineering Hierarchical NiMo/USY Catalysts for Selective Hydrocracking of Naphthalene to BTX

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Abstract

The selective hydrocracking of polycyclic aromatic hydrocarbons to BTX requires precise control over catalyst porosity and metal–acid balance. Hierarchical porosity, integrating microporous and mesoporous networks, is pivotal for enhancing mass transport and regulating reaction pathways. USY zeolites were engineered to create distinct hierarchical architectures via HCl, urea, and NaOH–surfactant treatments. HCl treatment constructed a gradient pore acidity system, urea treatment enhanced acidity while preserving microporosity, and NaOH–surfactant fabricated ordered mesopores with reduced acidity. The catalyst with the HCl-engineered gradient pore (NiMo/YH-1) achieved a 91% BTX yield at 425 °C in naphthalene hydrocracking, outperforming others. This performance is attributed to its gradient structure that enforces an optimal "hydrogenation-then-cracking" pathway, highlighting the critical role of tailored hierarchical porosity.

Keywords: USY zeolite; pore engineering; naphthalene; selective hydrocracking; BTX

1. Introduction

The global refining sector is undergoing a significant transition, driven by the "double carbon" goals and a steadily decreasing diesel-to-gasoline ratio. This shift necessitates a move from traditional fuel production toward higher-value chemical feedstocks, a strategy known as "oil reduction and chemical enhancement". Light cycle oil (LCO), a secondary stream from fluid catalytic cracking units, is a key target for upgrading due to its high aromatic content and poor diesel-blending properties [1–4]. The selective hydrocracking of LCO-derived polycyclic aromatic hydrocarbons (PAHs), such as naphthalene and phenanthrene, into benzene, toluene, and xylene (BTX) presents a promising route to simultaneously address diesel over-supply and meet the growing demand for basic petrochemicals [1–6].

Selective hydrocracking is a bifunctional process that requires a delicate balance between the hydrogenation function (typically provided by metals like Ni, Mo, W) and the cracking function (provided by solid acids) [5–7]. The reaction network for naphthalene, a model di-aromatic compound, involves initial hydrogenation to tetralin, followed by ring-opening and dealkylation to BTX [8]. An imbalance can lead to either over-hydrogenation to decalin or excessive cracking to light gases and coke. Beyond the acid–metal balance, molecular transport is critical [9–13]. The microporous structure of conventional zeolites

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like USY can impose diffusion limitations for bulky PAH molecules, leading to inefficient use of internal active sites and rapid deactivation [14,15].

Recent advances in the selective hydrocracking of polycyclic aromatic hydrocarbons PAHs BTX underscore the critical need for a precise metal-acid balance and efficient mass transport. This has been demonstrated across various catalyst systems, including the use of hybrid H-Beta/H-ZSM-5 zeolites with NiMo-S sites for tetralin conversion [6], non-noble Ni-Mo/HY–Al_2O_3 catalysts for the commercial upgrading of light cycle oil to high-octane gasoline [16], and NiW/Beta catalysts where nanoscale metal–acid proximity was key to enhancing BTX selectivity from 1-methylnaphthalene [17]. Parallel to these experimental efforts, detailed micro-kinetic modeling of biomass tar compounds over H-ZSM-5 has provided profound mechanistic insights into the complex reaction network leading to BTX [18]. Furthermore, pore-structure engineering of Y zeolites has been directly applied to naphthalene hydrocracking, where hierarchical Ni/Y catalysts with optimized acidity and metal dispersion achieved high BTX yields [19]. Collectively, these studies affirm the importance of tailored catalyst design. However, a systematic and comparative investigation into how fundamentally distinct post-synthetic methods (e.g., dealumination vs. desilication) engineer the hierarchical architecture of USY zeolites, and thereby dictate the spatial gradient of acidity and the sulfided metal phase, remains an outstanding challenge crucial for guiding rational catalyst design.

The construction of hierarchical pore systems, comprising interconnected microporous and mesoporous networks, is a crucial approach to improving mass transport efficiency and accessibility in zeolite materials [20–22]. Significant progress has been made in developing various postsynthetic methods for creating such hierarchical structures, each employing distinct mechanisms and yielding different outcomes. Beyond the well-established acid treatment that primarily removes extra-framework and framework aluminum to unblock and widen existing pores while modifying acidity [23,24], considerable attention has been directed toward alkaline-based strategies. Conventional alkaline treatment using NaOH selectively extracts silicon from the framework, creating intracrystalline mesopores and increasing acid site concentration through framework Al enrichment. To achieve better control over mesopore formation, more advanced approaches have emerged that combine alkalis with surfactants, guiding the desilication process toward more ordered and uniform mesoporous structures [25]. Recently, milder desilicating agents such as urea have also been explored as an alternative route for mesopore introduction with improved crystallinity preservation. Despite these developments, a systematic understanding of how these contrasting modification strategies comparatively influence the synergistic balance between porosity, acidity, and metal function in USY zeolites remains insufficiently explored, particularly for complex bifunctional catalysis like selective hydrocracking.

This study aims to provide a comprehensive understanding by engineering hierarchical USY pore systems via three contrasting methods: HCl treatment (targeting dealumination and pore cleaning), urea treatment (a mild desilication agent), and NaOH-surfactant treatment (for templated mesopore creation). We systematically correlate the engineered pore structures with the evolution of acidity, metal–support interaction, and the formation of the active phase, ultimately explaining their performance in the selective hydrocracking of naphthalene to BTX. Our findings underscore that the method of pore design dictates not only the physical transport pathways but also the chemical landscape of the catalyst, with profound implications for product selectivity.

2. Results and Discussion

2.1. Characterization of Supports

2.1.1. Structure Properties and Pore Structure of Supports

The X-ray diffraction (XRD) patterns of the parent and modified USY supports are presented in Figure 1a. All samples exhibit the characteristic diffraction peaks of the FAU topology, confirming that the crystalline framework was preserved after the various chemical treatments [26]. No additional peaks corresponding to impurity phases were detected, indicating the phase purity of the modified zeolites.

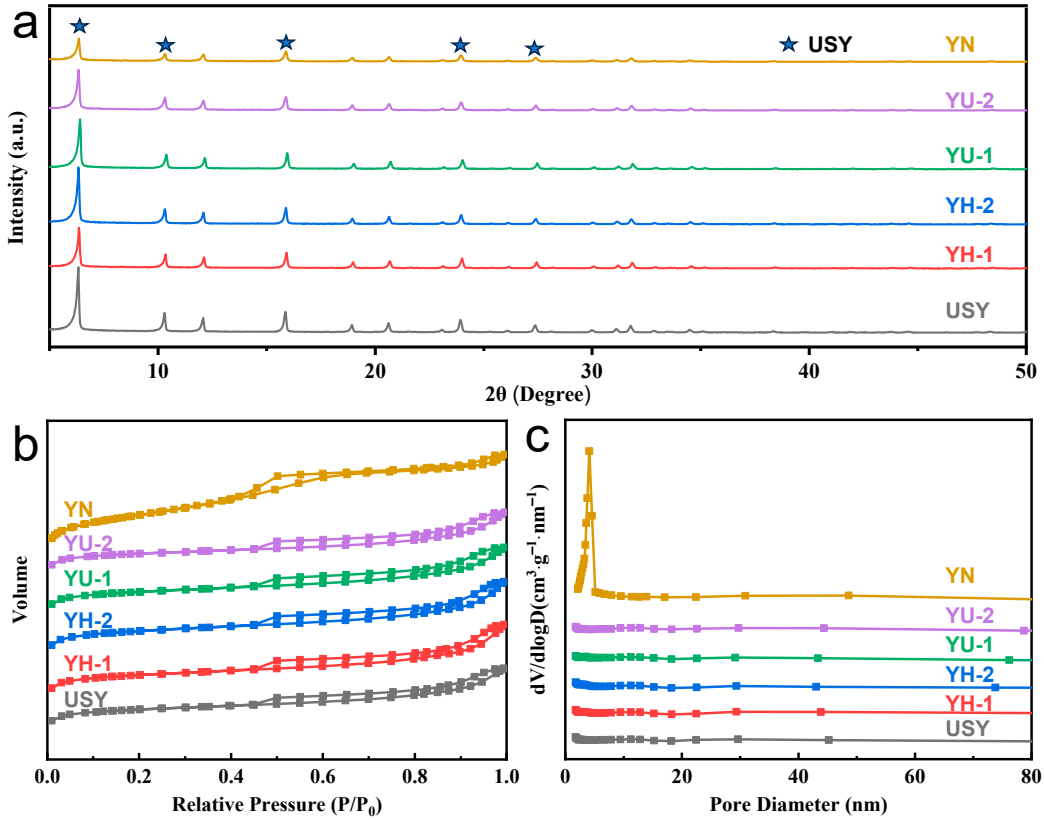

Figure 1. (**a**) wide-angle XRD patterns, (**b**) N_2 physisorption isotherms, (**c**) pore width distribution of supports.

A careful examination of the peak intensities and backgrounds reveals subtle differences. The relative crystallinity, estimated from the integrated intensity of the primary peaks in the 2θ range of 10–35°, decreased in the following order: USY ≈ YU-1 > YU-2 > YH-1 > YH-2 > YN. The hydrochloric acid-treated samples (YH-1, YH-2) showed a slight decrease in peak intensity, which can be attributed to the non-selective extraction of both framework and non-framework aluminum, causing some lattice disorder. The more pronounced decrease for YH-2 suggests that a higher acid concentration (0.3 M) compared to YH-1 (0.2 M) leads to a greater degree of dealumination and structural perturbation. The urea-treated samples (YU-1, YU-2) displayed minimal loss in crystallinity, consistent with the relatively mild nature of urea as a desilicating agent, which primarily attacks Si-rich regions without severely damaging the aluminosilicate framework. In stark contrast, the YN sample, modified with NaOH + CTAB, experienced the most significant reduction in crystallinity. This is a direct consequence of the aggressive desilication by NaOH, which dissolves a substantial portion of the silica framework, followed by a recrystallization process guided by the CTAB tem-

plate. While this creates new mesopores, it inevitably disrupts the long-range order of the original microporous crystal [22,25,27].

The N_2 adsorption–desorption isotherms and the corresponding pore size distribution (PSD) curves for the YX series are shown in Figure 1b,c, respectively, with the quantitative textual parameters summarized in Table 1. The parent USY zeolite exhibits a Type IV isotherm with a pronounced H4-type hysteresis loop, which is typical for micro-mesoporous materials with slit-shaped pores arising from the aggregation of plate-like particles [10]. The presence of a steep nitrogen uptake at low relative pressures ($P/P_0 < 0.01$) indicates a well-developed microporosity, while the hysteresis loop at higher pressures confirms the existence of mesopores, likely the intrinsic secondary pores of USY.

Table 1. Pore Structure Parameters of YX Series Supports.

Samples	Pore Size (nm)	V_{total} (cm^3/g)	V_{mic} (cm^3/g)	V_{mes} (cm^3/g)	S_{BET} (m^2/g)	S_{mic} (m^2/g)	S_{ext} (m^2/g)	SiO_2/Al_2O_3 (Molar)
USY	7.2	0.43	0.27	0.16	710	578	132	5.5
YH-1	7.5	0.45	0.26	0.19	709	554	155	7.2
YH-2	7.1	0.45	0.26	0.19	719	553	166	9.5
YU-1	6.9	0.42	0.25	0.17	690	545	145	5.1
YU-2	7.4	0.41	0.25	0.16	674	540	134	4.7
YN	4.2	0.53	0.22	0.31	744	494	250	4.0

The PSD curves (Figure 1c) indicate that the acid treatment broadened the existing mesopores centered around 7–8 nm rather than creating new, distinct mesopores. This is attributed to the dissolution of non-framework alumina species that partially block the pore entrances and channels, thereby "cleaning" and interconnecting the intrinsic pore system [23,24]. As seen in Table 1, this led to a slight increase in the total pore volume (V_{total}) and a more noticeable increase in the mesopore volume (V_{mes}) and external surface area (S_{ext}), at the expense of a small decrease in the micropore volume (V_{mic}) and micropore surface area (S_{mic}). The dealumination process creates voids within the crystals, contributing to the mesopore volume. The comparable textual parameters of YH-1 and YH-2 suggest that increasing the HCl concentration from 0.2 M to 0.3 M has a limited additional impact on pore structure development under the conditions used.

The urea-treated samples (YU-1, YU-2) also displayed similar isotherms to USY. The desilication effect of urea is less aggressive than that of NaOH. It primarily creates small, spot-like mesopores within the crystals, as evidenced by a slight increase in S_{ext} and a subtle broadening of the PSD towards smaller mesopore sizes (~2–5 nm). The decrease in S_{BET} and V_{mic} for YU-1 and YU-2 indicates that some microporous structure is lost during the mild silicon extraction process.

The YN sample, modified with NaOH + CTAB, displayed a dramatically different isotherm. It showed a much larger nitrogen uptake in the relative pressure range of 0.4–0.9, characteristic of capillary condensation within well-defined mesopores. The hysteresis loop remains of H4 type but is much more pronounced. The PSD curve (Figure 1c) reveals a sharp, uniform peak centered at approximately 4.2 nm, unequivocally demonstrating the successful introduction of ordered mesopores within the hierarchical structure via the surfactant-templated recrystallization process [27,28]. This is further supported by the textual data in Table 1: YN possesses the highest total pore volume (0.53 cm^3/g) and the largest external surface area (250 m^2/g) among all samples, confirming the successful creation of a hierarchical structure with significant mesoporosity. However, this comes at the cost of a significant reduction in micropore volume (0.22 cm^3/g) and micropore surface area (494 m^2/g), indicating a restructuring rather than a complete loss of the microporous framework.

2.1.2. Microstructure and Morphology

TEM images confirmed the preservation of the intrinsic zeolite pore structure and morphology (slit-shaped pores, secondary pores) across all supports. TEM images provided direct visual evidence of the morphological changes induced by the different modifications. For the hydrochloric acid-treated samples YH-1 and YH-2 (Figure 2b,c), the images show that the fundamental zeolite morphology is retained. The most notable change is that the existing slit-shaped mesopores and secondary pores appear more "etched" and clearer, with enhanced contrast. This is consistent with the N_2 physisorption results and confirms that the acid treatment acts as a "pore-cleaning" process, removing amorphous debris and EFAL to open up the pre-existing pore network without drastically altering the overall particle morphology [23,24]. The urea-treated samples YU-1 and YU-2 (Figure 2d,e) display a similar overall morphology to USY. However, upon closer inspection, one can observe the presence of numerous small, dark, spot-like features within the crystals, which are not as prevalent in the parent USY. These features are attributed to the small mesopores created by the selective extraction of silicon. With increasing urea concentration (YU-2), these spot-like mesopores become more numerous and slightly larger, indicating a more pronounced desilication effect. The YN sample (Figure 2f) presents a markedly different hierarchical architecture. The dense microporous framework is largely replaced by a worm-like or sponge-like network of interconnected mesopores with high uniformity in size, consistent with the sharp PSD peak at 4.2 nm. This image visually corroborates the formation of a composite hierarchical porosity at the expense of some microporosity, as quantified in Table 1 [27,29].

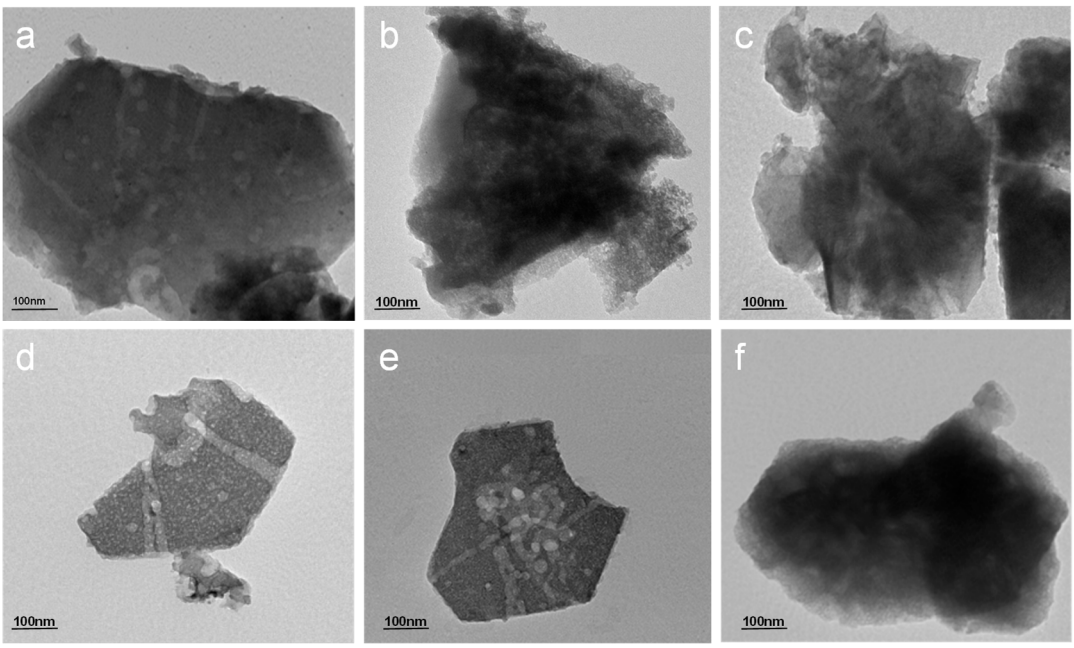

Figure 2. TEM images of YX series supports: (**a**) USY; (**b**) YH-1; (**c**) YH-2; (**d**) YU-1; (**e**) YU-2; (**f**) YN.

2.2. Characterization of NiMo/YX Catalysts

2.2.1. Structure Properties and Pore Structure of Catalysts

The XRD patterns of the sulfided NiMo/YX catalysts are shown in Figure 3. All patterns are dominated by the diffraction peaks of the USY framework, which are largely consistent with those of their corresponding supports (Figure 1a). This indicates that the impregnation and subsequent calcination/sulfidation processes did not cause further significant damage to the zeolite crystal structure. Crucially, no distinct diffraction peaks corresponding to crystalline MoO_3 (e.g., at $2\theta = 23.3°, 25.7°, 27.3°$) or NiO (e.g., at

2θ = 37.3°, 43.3°) are observed in any of the catalysts [15,29]. This suggests that the active metal precursors are well-dispersed on the surface of the modified USY supports, likely forming small oxide clusters or monolayers that are below the detection limit of XRD (~4–5 nm). The absence of sharp metal oxide peaks is a positive indicator of a high dispersion of the active metal phases, which is essential for creating a high density of active sites.

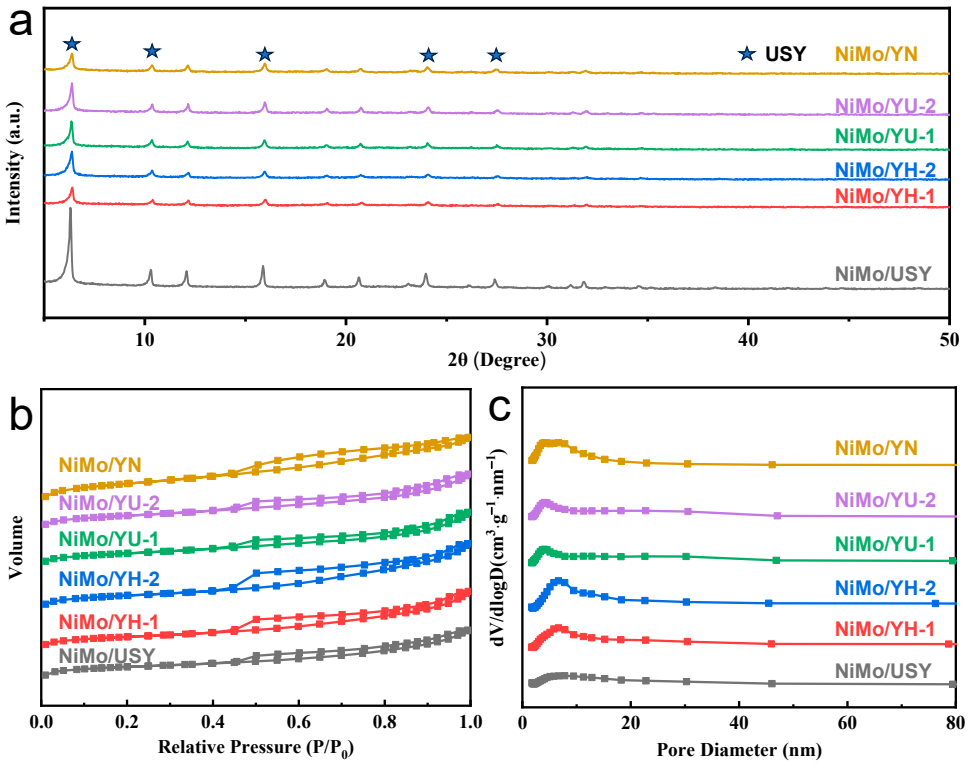

Figure 3. (**a**) wide-angle XRD patterns, (**b**) N_2 physisorption isotherms, (**c**) pore width distribution of catalysts.

The N_2 physisorption isotherms and PSD curves of the NiMo/YX catalysts are presented in Figure 3b,c, with the corresponding textual parameters listed in Table 2. A comparison with Table 1 reveals that loading NiMo species resulted in a universal decrease in surface area and pore volume for all catalysts, which is expected as the metal phases occupy a portion of the pore space.

Table 2. Pore Structure Parameters of NiMo/YX Series Supports.

Samples NiMo/	Pore Size (nm)	V_{total} (cm^3/g)	V_{mic} (cm^3/g)	V_{mes} (cm^3/g)	S_{BET} (m^2/g)	S_{mic} (m^2/g)	S_{ext} (m^2/g)	SiO_2/Al_2O_3 (Molar)	NiO (wt%)	MoO_3 (wt%)
USY	6.8	0.27	0.14	0.13	383	296	87	5.5	1.85	11.2
YH-1	6.3	0.24	0.07	0.17	244	166	78	7.2	1.83	11.1
YH-2	5.8	0.27	0.07	0.20	258	170	88	9.5	1.82	11.0
YU-1	6.5	0.24	0.09	0.15	277	199	78	5.1	1.86	11.3
YU-2	6.3	0.26	0.11	0.15	333	244	89	4.7	1.87	11.3
YN	5.2	0.30	0.11	0.19	386	258	128	4.0	1.81	11.0

The NiMo/USY catalyst experienced a significant reduction in S_{BET} (from 710 to 383 m^2/g) and V_{mic} (from 0.27 to 0.14 cm^3/g), indicating that a substantial amount of metal has entered and potentially blocked the micropores. For the treated catalysts, the trend is similar but reveals important differences. The NiMo/YH-1 and NiMo/YH-2 catalysts show a very low residual micropore volume (0.07 cm^3/g) and micropore surface area (~170 m^2/g). This suggests that the "cleaned" and interconnected pore network of

the acid-treated supports allowed for efficient diffusion of the metal precursors during impregnation, leading to a relatively uniform deposition of metals that also covered a large portion of the micropore surface.

Notably, NiMo/YN, utilizing the support with surfactant-templated mesopores, experienced the smallest relative decrease in S_{BET} among the modified catalysts, retaining the highest overall surface area (386 m^2/g). Its external surface area (S_{ext} = 128 m^2/g) is also the highest of all catalysts after metal loading. This can be explained by the unique hierarchical pore structure of YN. The large, uniform mesopores introduced by the template method are less prone to being completely blocked by the metal particles, while the remaining micropores contribute to the preserved surface area. Instead, the metals are likely dispersed on the walls of these spacious mesopores, preserving a significant portion of the surface area and leaving the remaining micropores more accessible than in other samples. This hierarchical architecture is highly beneficial for mass transport.

2.2.2. Surface Acidity Properties of NiMo/YX Catalysts

The acid properties of the catalysts, crucial for the cracking function, were characterized by NH_3-TPD and Py-FTIR (Figure 4, Table 3). NH_3-TPD profiles showed two distinct desorption peaks: a low-temperature peak (150–300 °C) corresponding to weak acid sites and a high-temperature peak (350–550 °C) representing strong acid sites [29]. The parent NiMo/USY exhibited a broad, intense high-temperature peak, indicating abundant strong acid sites. In contrast, hydrochloric acid-treated catalysts (NiMo/YH-1, YH-2) showed significantly reduced peak areas, especially for YH-2, due to dealumination decreasing Brønsted acid sites. Conversely, urea-treated catalysts (NiMo/YU-1, YU-2) displayed dramatically enhanced peak areas from desilication increasing framework Al density. The NiMo/YN catalyst showed a dominant low-temperature peak but minimal high-temperature desorption, indicating substantial loss of strong acidity from aggressive NaOH treatment.

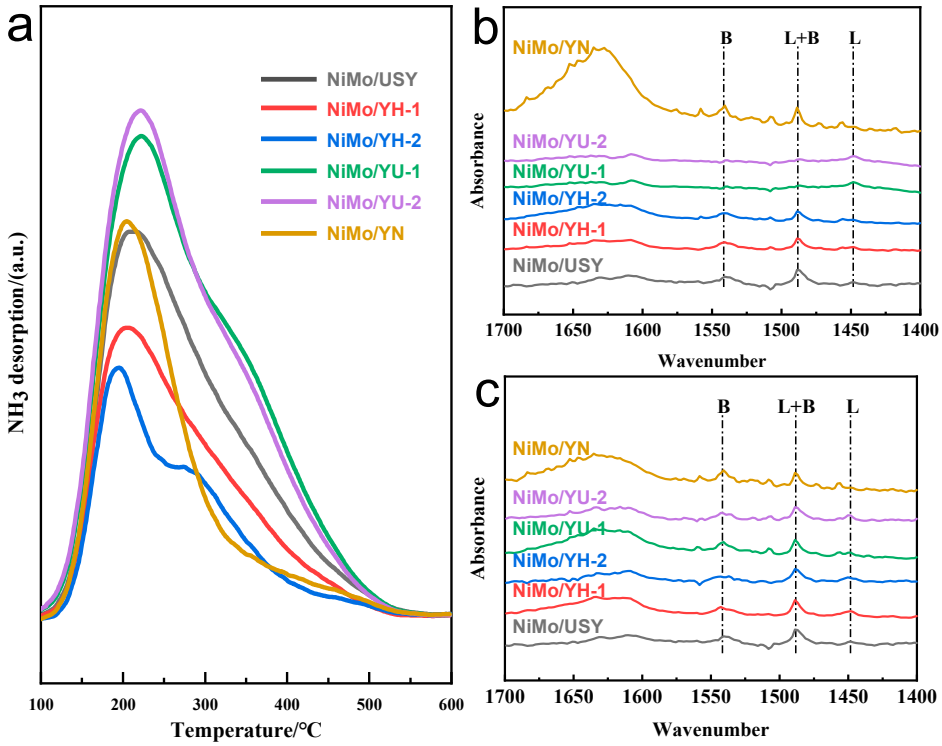

Figure 4. (**a**) NH3-TPD spectra of catalysts, (**b**)Py-IR spectra after desorbing at 200 °C, (**c**) Py-IR spectra after desorbing at 350 °C.

Table 3. Acid Properties of NiMo/YX Series Catalysts.

Samples NiMo/	Total Acid μmol/g	Weak Acid, μmol/g [a]			Strong Acid, μmol/g [a]			Acid Sites, μmol/g [b]			H_2 Consumption, μmol/g [c]		
		L	B	L + B	L	B	L + B	Weak	Strong	Total	420–460 °C	500–550 °C	700–750 °C
USY	870.4	102.2	230.9	333.1	102.2	428.3	530.5	391.4	708.1	1108	668.1	194.2	391.1
YH-1	619.2	72.0	158.1	230.1	165.7	223.4	389.1	304.6	501.4	806	570.9	451.6	247.5
YH-2	450.4	115.6	40.4	156.0	55.9	238.5	294.4	203.0	364.0	559	476.6	352.5	402.9
YU-1	1116.5	211.5	162.3	373.8	211.9	530.8	742.7	469.3	895.4	1364.6	549.3	384.5	427.2
YU-2	1172.4	134.3	258.2	392.5	197.7	582.2	779.9	475.6	1014.9	1490.5	571.0	628.7	440.2
YN	629.1	166.7	250.2	416.9	50.8	161.4	212.2	538.1	271.3	809.4	706.1	557.7	220.2

[a] The amount and property of acid was analyzed by Py-IR. [b] The amount and property of acid was analyzed by NH_3-TPD. [c] The amount of H2 consumption was analyzed by H2-TPR.

Py-FTIR analysis provided detailed acid-type quantification [30–32]. NiMo/USY maintained balanced Brønsted/Lewis acid distribution with substantial strong acidity. HCl treatment preferentially reduced weak Brønsted and strong Lewis acids, creating an acid strength gradient from crystal exterior to interior. Urea modification drastically enhanced both Brønsted and Lewis acidity, particularly strong Brønsted acids effective for C-C scission but prone to coking. The YN catalyst retained moderate total acidity but exhibited the lowest strong acid density, rendering it less effective for ring-opening reactions. These distinct acid profiles directly correlate with the observed catalytic performances in naphthalene hydrocracking.

2.2.3. Metal–Support Interaction and Reducibility

H_2-TPR was used to investigate the reducibility of the oxide precursors and the metal–support interaction (MSI), which influences the sulfidation behavior and final active phase morphology. The profiles are shown in Figure 5. The NiMo/USY catalyst shows a main reduction peak around 450–500 °C, attributed to the reduction in octahedrally coordinated Mo^{6+} species to Mo^{4+}, which is a precursor for the formation of the active MoS_2 phase. A smaller, higher-temperature peak above 700 °C is associated with the reduction in tetrahedral Mo species or strongly interacting MoOx clusters [33–35].

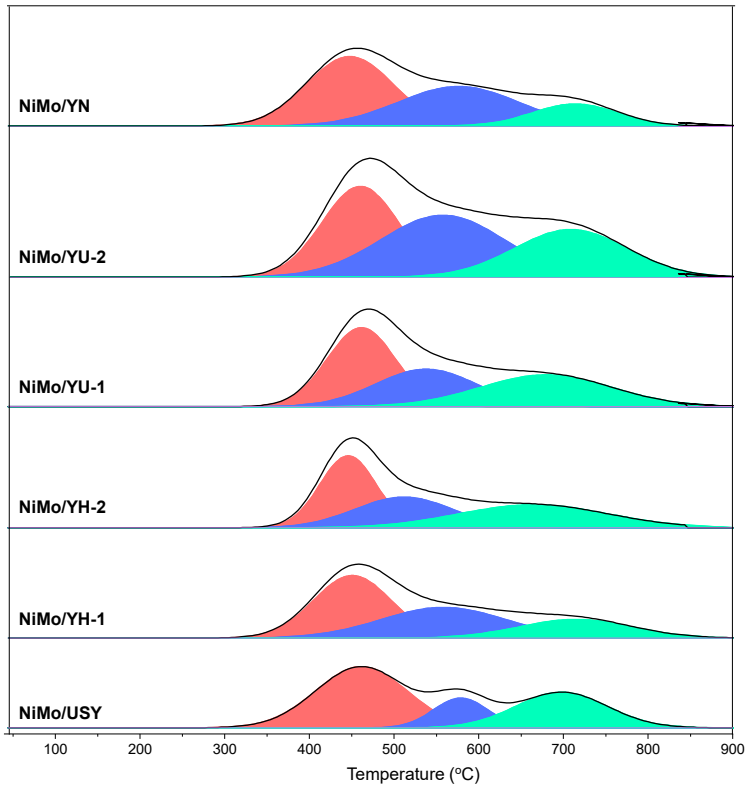

Figure 5. H_2-TPR characterization of NiMo/YX series catalysts.

For the hydrochloric acid-treated catalysts (NiMo/YH-1, NiMo/YH-2), the main reduction peak shifts to a lower temperature (~420–460 °C), and its intensity increases. The high-temperature reduction peak diminishes. This indicates that the dealumination of the USY support weakens the interaction between the Mo species and the support surface. A weaker MSI makes the MoO_3 particles easier to reduce and, consequently, easier to sulfide, which is generally beneficial for creating the active NiMoS phase. In contrast, the urea-treated catalysts (NiMo/YU-1, NiMo/YU-2) show a shift in the main reduction peak to higher temperatures (~470–500 °C). The increased aluminum content and stronger acidity of these samples strengthen the MSI, making the reduction in Mo^{6+} more difficult. This stronger interaction can hinder the mobility and sulfidation of the metal species. The NiMo/YN catalyst exhibits a complex reduction profile with a broad, low-temperature peak. The significant decrease in strong acidity and the presence of new mesoporous silica-alumina walls after recrystallization create a different chemical environment for the metals, leading to a weaker MSI similar to the acid-treated samples. However, the overall lower reducibility and different profile shape suggest a different dispersion and coordination of the metal species.

2.2.4. Chemical State Analysis of Active Phase

XPS analysis of the sulfided catalysts provides direct information about the chemical state of Mo and the degree of sulfidation, which is critical for hydrogenation activity. The deconvoluted Mo 3d spectra are shown in Figure 6, and the quantitative distribution of Mo species is summarized in Table 4. The Mo 3d spectrum can be fitted into three doublets: Mo^{4+} (229.0–229.3 eV for Mo $3d_5/_2$), corresponding to MoS_2; Mo^{5+} (230.5–231.0 eV), an intermediate oxy-sulfide species; and Mo^{6+} (232.4–232.8 eV), corresponding to unsulfided MoO_3 [36–39]. The proportion of Mo^{4+} is a key indicator of a successful sulfidation process. NiMo/USY has a relatively high sulfided Mo^{4+} content of 58.8%. The hydrochloric acid-treated catalysts show a decrease in Mo^{4+}, with NiMo/YH-2 having a very low value of 35.8%. This is counterintuitive given the weaker MSI observed in H_2-TPR, which should favor sulfidation. A possible explanation is that the severe dealumination significantly reduces the number of anchoring sites (e.g., Al-OH groups) for the Mo precursors, leading to the formation of larger, less sulfidable MoO_3 aggregates during calcination. Alternatively, the reduced acidity might itself play a role in the sulfidation thermodynamics or kinetics.

The urea-treated catalysts maintained a high Mo^{4+} content (~54–57%), similar to the parent catalyst, despite the stronger MSI. This suggests that the high acid density in these samples provides sufficient anchoring sites and may facilitate the sulfidation process through proximity to protons. NiMo/YN exhibits the second-lowest Mo^{4+} content (35.6%). The combination of low acidity and a restructured, silica-rich surface in YN appears to be highly detrimental to the sulfidation of Mo species, leaving a large fraction in an oxidized state (Mo^{6+} and Mo^{5+}). This poor sulfidation directly implies a lower density of active hydrogenation sites.

Table 4. Mo(W) XPS peak data of molybdenum (tungsten) species in sulfided NiMo/YX series catalysts.

Samples NiMo/	$Mo(W)^{4+}$ %	$Mo(W)^{5+}$ %	$Mo(W)^{6+}$ %
USY	58.8	6.2	35.0
YH-1	48.8	7.1	44.1
YH-2	35.8	48.5	15.7
YU-1	53.9	21.0	25.1
YU-2	56.7	18.1	25.2
YN	35.6	33.0	31.4

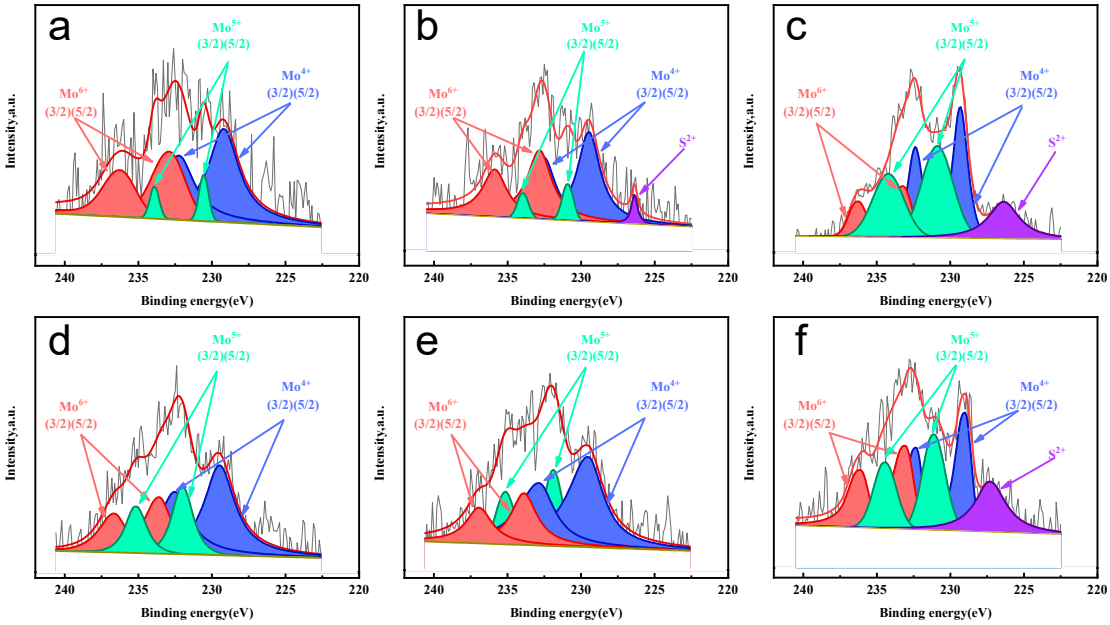

Figure 6. Mo(W) XPS spectra of sulfided NiMo/YX series catalysts: (**a**) NiMo/USY; (**b**) NiMo/YH-1; (**c**) NiMo/YH-2; (**d**) NiMo/YU-1; (**e**) NiMo/YU-2; (**f**) NiMo/YN.

2.2.5. Morphology of the MoS_2 Slabs

High-resolution TEM (HRTEM) was employed to visualize the morphology of the active MoS_2 phase. Representative images and the statistical analysis of slab length and layer number are shown in Figure 7, Figure 8 and Table 5, respectively. The active phase in hydrocracking catalysts typically consists of stacked MoS_2 slabs promoted by Ni. The average slab length (L_{av}) is related to the dispersion of the active phase, while the average number of stacking layers (N_{av}) is influenced by the metal–support interaction and promoter content [36]. The NiMo/USY catalyst shows well-dispersed MoS_2 slabs with an average length of 4.6 nm and 3.6 layers per stack. The acid-treated catalysts (NiMo/YH-1, YH-2) show a slight decrease in slab length (to 4.0 and 3.8 nm, respectively) but a slight increase in stacking (to 3.8 and 4.0 layers). The weaker MSI may allow for a greater mobility of MoS_2 precursors during sulfidation, facilitating stacking over lateral growth. The urea-treated catalysts (NiMo/YU-1, YU-2) displayed a slight increase in slab length and a decrease in stacking. The stronger MSI in these samples may pin the MoS_2 slabs, limiting their mobility and thus hindering both lateral growth and stacking [40]. The NiMo/YN catalyst showed relatively long slabs (4.4 nm) and high stacking (3.9 layers). The poor sulfidation and unique support environment seem to lead to the formation of larger, more aggregated MoS_2 crystallites.

Table 5. Average length (L_{av}) and average layer number (N_{av}) of MoS_2 crystallites.

Samples NiMo/	L_{av} (nm)	N_{av}	f_{Mo}
USY	4.6	3.6	0.21
YH-1	4.0	3.8	0.23
YH-2	3.8	4.0	0.24
YU-1	4.2	3.2	0.22
YU-2	4.3	3.1	0.22
YN	4.4	3.9	0.21

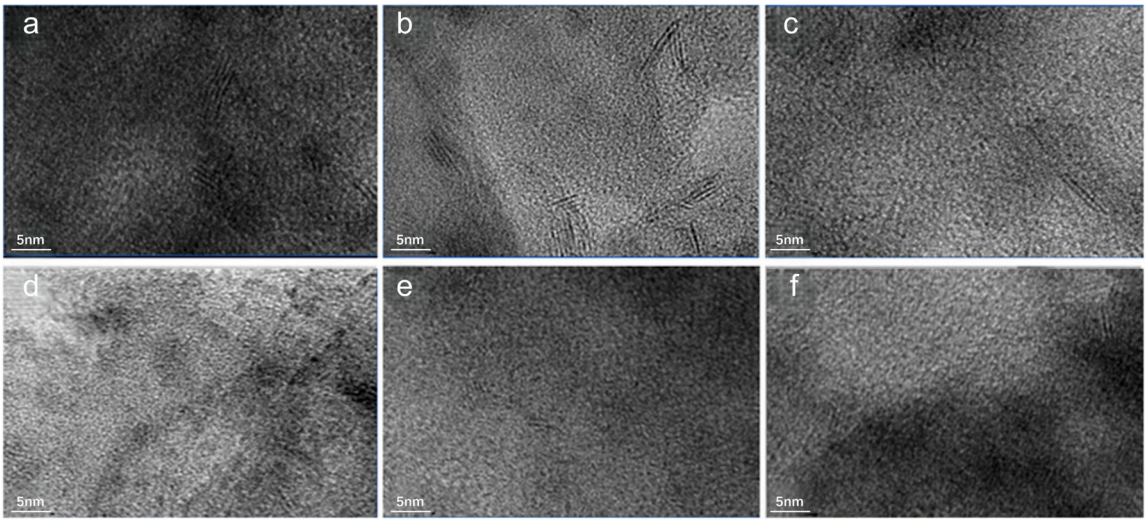

Figure 7. HRTEM image of NiMo/YX series catalysts: (**a**) NiMo/USY; (**b**) NiMo/YH-1; (**c**) NiMo/YH-2; (**d**) NiMo/YU-1; (**e**) NiMo/YU-2; (**f**) NiMo/YN.

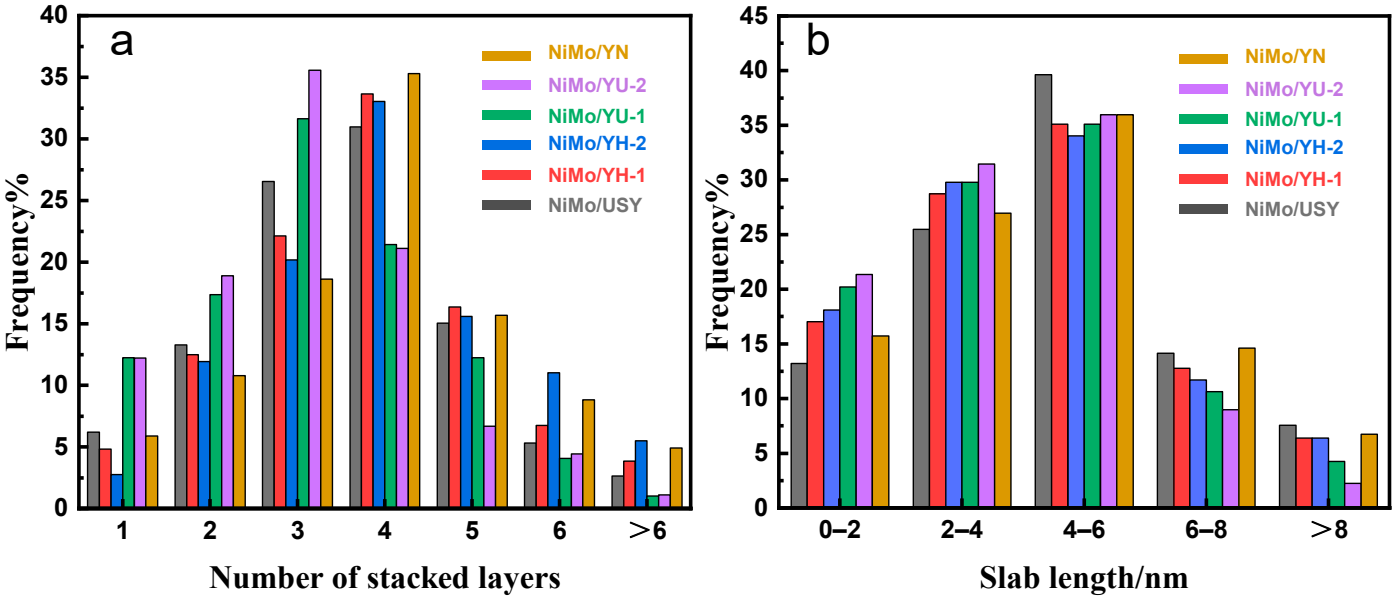

Figure 8. Distribution of Stacking Layers (**a**) and Stacking Length (**b**) of NiMo/YX Series Catalysts.

2.3. Hydrocracking Performance of NiMo/YX Catalysts

The catalytic performance of the NiMo/YX catalysts was evaluated in the selective hydrocracking of naphthalene from 350 to 425 °C. Compared with the literature [10,28,41–45], the catalyst synthesized in this work exhibits superior performance, A detailed comparison is presented in Table S1. The naphthalene conversion and the yield of the desired benzene, toluene, and xylenes (BTX) are shown in Figure 9. The detailed product distribution is shown in Figure S1.

Analysis of naphthalene conversion revealed that most catalysts achieved high activity (>94%) in the range of 350–400 °C, with the urea-treated catalysts (NiMo/YU-1, YU-2) and NiMo/YH-1 demonstrating particularly robust performance. A notable decline in conversion was observed at 425 °C for the parent NiMo/USY and the NiMo/YN catalyst, indicating their structures were less effective under the most severe conditions. The yield of BTX, however, provides critical insights for evaluating selective hydrocracking performance. A pivotal finding is that the NiMo/YH-1 catalyst achieved a superior BTX yield of 91% at 425 °C, the highest among all catalysts. This outstanding performance is

particularly notable as it was sustained at the highest temperature, while other catalysts, notably NiMo/USY, experienced a significant decline. The BTX yield trends further highlight the structural advantages: the NiMo/YH-1 and NiMo/YH-2 catalysts exhibited a consistent and sharp increase in BTX yield with temperature, culminating in the maximum yield for YH-1. In contrast, the highly active NiMo/YU-series catalysts reached a BTX yield plateau of approximately 80–85%, suggesting a limitation in ultimate selectivity despite high conversion, while the NiMo/YN catalyst produced negligible BTX yields at all temperatures, consistent with its inherent functional deficiencies. Thermogravimetric analysis further revealed that the spent NiMo/YH-1 catalyst had significantly less carbon deposition (Figure S2), thus demonstrating its superior hydrocracking performance. These results decisively demonstrate that the exceptional and sustained BTX yield of NiMo/YH-1 is the key outcome, underscoring the critical advantage conferred by its specific hierarchical gradient architecture, which will be mechanistically elucidated in the following section.

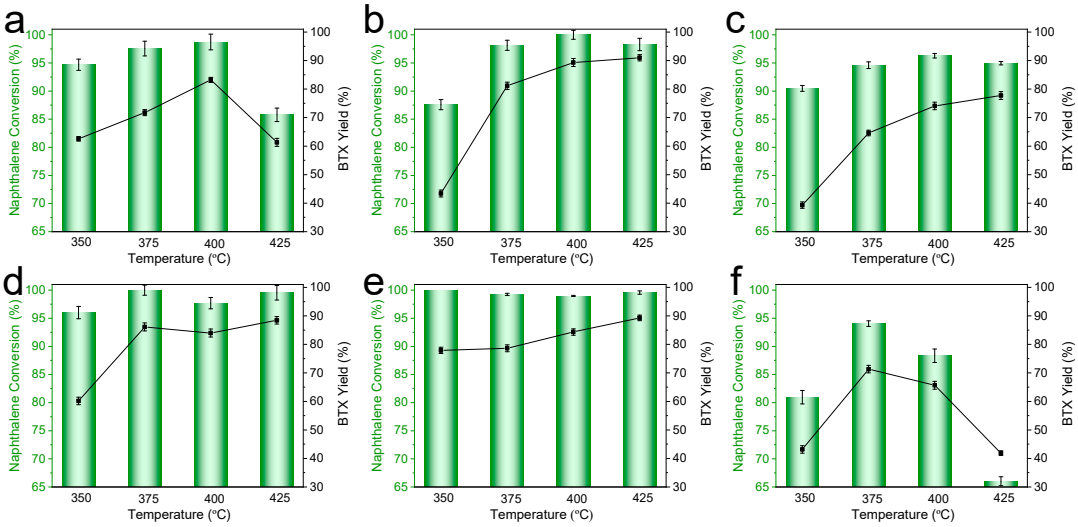

Figure 9. Conversion and BTX yield of naphthalene hydrocracking at different reaction temperature over (**a**) NiMo/USY; (**b**) NiMo/YH-1; (**c**) NiMo/YH-2; (**d**) NiMo/YU-1; (**e**) NiMo/YU-2; (**f**) NiMo/YN. ((Reaction conditions: T = 350–425 °C, P = 4.0 MPa, H_2/oil = 600 (volume ratio) and WHSV = 4 h^{-1}; and feedstock composition (5 wt% naphthalene in n-octane)).

2.4. Relationships Catalytic Performance in Naphthalene Hydrocracking

The marked differences in catalytic performance, particularly the superior BTX yield achieved by NiMo/YH-1, can be rationalized by synthesizing the physiochemical characterizations with the reaction data (Figure 10). The distinct post-synthetic treatments engineered not only the porosity but also the acidic and metallic landscapes of the USY zeolites, which collectively dictated the reaction pathway and product distribution.

The exceptional performance of the NiMo/YH-1 catalyst is attributed to its unique gradient hierarchical pore acidity system constructed by HCl treatment. This treatment preferentially dealuminates the more accessible external surface and mesopores, creating a structure where the acid strength increases from the crystal exterior to the microporous core. After metal loading, this results in a spatial differentiation of functions: the mesopores are rich in hydrogenation/dehydrogenation metals (NiMoS) but lean in strong acid sites, while the micropores retain a higher concentration of strong acid sites. This specific architecture enforces an ideal "hydrogenation-then-cracking" sequence. Bulky naphthalene molecules first access the metal-rich, large mesopores where they are selectively hydrogenated to tetralin. This intermediate then diffuses into the acidic micropores for selective ring-opening and dealkylation into BTX. This spatial separation is crucial as it prevents naphthalene from directly contacting strong acid sites, thereby minimizing polymerization and coke

formation, and avoids over-hydrogenation of tetralin to decalin [15]. Furthermore, the acid-cleaned and interconnected pore network facilitates rapid diffusion, reducing the residence time of intermediates and suppressing secondary reactions [23,24]. The H_2-TPR profile confirms a weakened metal–support interaction which favors sulfidation, while HRTEM shows well-dispersed MoS_2 slabs, ensuring adequate hydrogenation functionality.

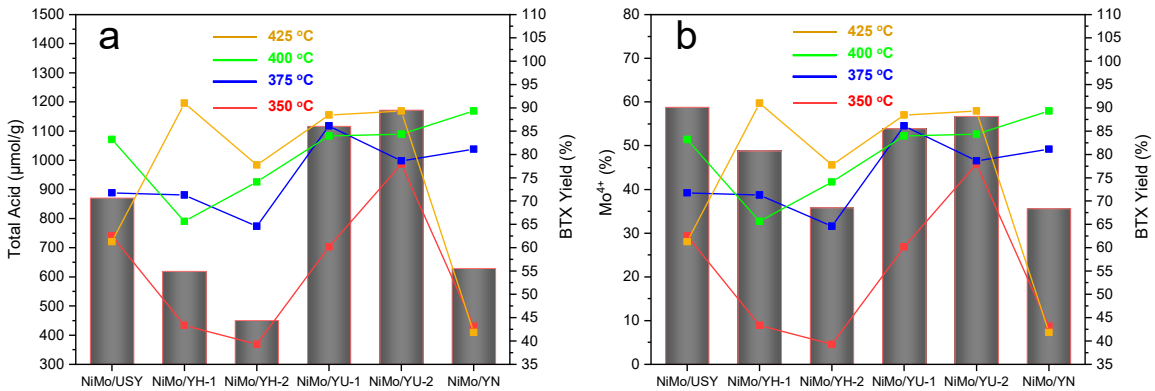

Figure 10. (**a**) Total Strong Acidity (from Py-IR) vs. BTX Yield, and (**b**) $Mo^{4+}/(Mo^{4+} + Mo^{5+} + Mo^{6+})$ Ratio (from XPS, representing sulfidation degree) vs. BTX Yield from 350 °C to 425 °C for all NiMo/YX catalysts.

In contrast, the NiMo/YU-series catalysts, despite their high acidity and near-complete naphthalene conversion, reached a BTX yield plateau. This limitation stems from their uniformly distributed, very high density of strong acid sites within a non-gradient hierarchical framework. The lack of spatial control leads to parallel reaction pathways where naphthalene can undergo direct, non-selective cracking on strong acid sites, leading to light gases, and primary BTX products can be over-cracked. Although their hydrogenation function is competent, it cannot fully compensate for the indiscriminate cracking activity, resulting in high activity but compromised ultimate BTX selectivity [22,25,27].

The poor performance of the NiMo/YN catalyst stems from a fatal imbalance between its metal and acid functions. The aggressive NaOH-surfactant treatment successfully created ordered mesopores but simultaneously drastically reduced the density of strong acid sites and left residual sodium that further neutralizes acidity. Concurrently, the restructured, silica-rich surface proved highly detrimental to metal sulfidation, as evidenced by the low Mo^{4+} content in XPS analysis. Consequently, NiMo/YN possesses neither a sufficient number of strong acid sites for effective cracking nor an adequate density of sulfided hydrogenation sites, failing to catalyze either step of the bifunctional mechanism efficiently. The underperformance of the parent NiMo/USY catalyst, especially at 425 °C, highlights the limitations of a conventional microporous-dominated structure. Its non-gradient acidity and potential diffusion limitations for bulky reactants and intermediates at high reaction severity make it prone to undesired secondary reactions, leading to a significant drop in both conversion and BTX yield.

3. Experimental Procedure

3.1. Materials

The USY zeolite with a silica-to-alumina ratio (SiO_2/Al_2O_3) of 5.5 was supplied by Sinopec Catalyst Co., Ltd. (Beijing, China). The modifying agents, including hydrochloric acid (HCl, 36–38%), urea ($CO(NH_2)_2$, ≥99.0%), sodium hydroxide (NaOH, ≥96.0%), and the surfactant cetyltrimethylammonium bromide (CTAB, ≥99.0%), were all of analytical grade and purchased from Shanghai Aladdin Biochemical Technology Co., Ltd.(Shanghai, China) The metal precursors, ammonium heptamolybdate tetrahydrate ($(NH_4)_6Mo_7O_{24}\cdot 4H_2O$, ≥99.0%) and nickel nitrate hexahydrate ($Ni(NO_3)_2\cdot 6H_2O$, ≥98.0%), were also obtained

from Shanghai Aladdin Biochemical Technology Co., Ltd. (Shanghai, China). Deionized water was used throughout the experiments for all washing and solution preparation steps. All chemicals were utilized as received without any further purification.

3.2. Design of USY Pore Architecture via Postsynthetic Engineering

Three distinct chemical treatment routes were employed to engineer the pore structure of the parent USY zeolite, each targeting a specific type of pore architecture. Three distinct chemical treatment routes were employed to engineer the pore structure of the parent USY zeolite, each targeting a specific type of pore architecture. The procedures for all synthesized supports are summarized in Table 6.

Table 6. Summary of USY support modification methods.

Sample	Treatment Method	Treatment Conditions (Concentration, Temperature, Time)
USY	Parent material	
YH-1	Hydrochloric acid treatment	0.2 M HCl, 80 °C, 4 h
YH-2	Hydrochloric acid treatment	0.3 M HCl, 80 °C, 4 h
YU-1	Urea treatment	1.0 M Urea, 80 °C, 4 h
YU-2	Urea treatment	1.5 M Urea, 80 °C, 4 h
YN	NaOH-Surfactant treatment	0.2 M NaOH + CTAB, 150 °C, 24 h (hydrothermal)

3.2.1. Hydrochloric Acid Treatment (Gradient Pore System)

This treatment aimed to create a gradient pore system by selective dealumination. In a typical procedure, 10.0 g of pristine USY zeolite was dispersed into 100 mL of an HCl solution (0.2 mol/L or 0.3 mol/L) in a 250 mL round-bottom flask. The solid-to-liquid ratio was maintained at 1:10 (g:mL). The mixture was then vigorously stirred and heated in a water bath at a constant temperature of 80 °C for a duration of 4 h. After the reaction, the solid product was recovered by vacuum filtration and thoroughly washed with copious amounts of deionized water until the pH of the filtrate became neutral. The resulting filter cake was dried in an oven at 110 °C for 12 h. Finally, the dried powder was calcined in a muffle furnace at 550 °C for 4 h under static air to obtain the final modified supports, designated as YH-1 (from 0.2 M HCl) and YH-2 (from 0.3 M HCl).

3.2.2. Urea Treatment (Mild Mesopore Introduction)

This method utilized mild desilication to introduce mesoporosity while largely preserving the framework. The procedure was similar to the acid treatment. Specifically, 10.0 g of USY was treated with 100 mL of a urea solution (1.0 mol/L or 1.5 mol/L) at 80 °C for 4 h under stirring. The subsequent steps of filtration, washing until neutral, drying (110 °C, 12 h), and calcination (550 °C, 4 h) were identical. The samples obtained from this treatment were labeled as YU-1 (1.0 M urea) and YU-2 (1.5 M urea).

3.2.3. NaOH-Surfactant Treatment (Ordered Mesopore Creation)

This two-step process was designed to create a more uniform and ordered mesoporous structure. First, 10.0 g of USY was mixed with 100 mL of a 0.2 mol/L NaOH solution and stirred at 80 °C for 2 h to initiate desilication. Subsequently, 1.0 g of the structure-directing agent CTAB was added to the mixture, and stirring continued for another 30 min to allow the surfactant to interact with the zeolite surface. The entire slurry was then transferred into a Teflon-lined stainless-steel autoclave and subjected to hydrothermal crystallization at 150 °C for 24 h. The resulting product was filtered, washed, dried, and calcined as described above. To remove residual sodium ions and convert the material into its acidic form, the calcined product underwent two consecutive ion-exchange cycles with a 1.0 mol/L NH_4NO_3 solution (solid-to-liquid ratio of 1:10 g/mL, 80 °C, 2 h per cycle), each

followed by drying. A final calcination step at 550 °C for 4 h was performed to yield the NH_4^+-exchanged and then H^+-form zeolite, designated as YN.

3.3. Catalyst Preparation via Sequential Impregnation

The NiMo catalysts were prepared using a sequential incipient wetness impregnation method onto the modified USY supports (collectively referred to as YX). The target total loading of the metal oxides (MoO_3 + NiO) was fixed at 18.5 wt% for all catalysts. In the first step, a calculated amount of ammonium heptamolybdate tetrahydrate, sufficient to achieve a 16.5 wt% MoO_3 loading, was dissolved in a volume of deionized water precisely equal to the total pore volume of the support. This solution was then added dropwise to the YX support under continuous manual stirring to ensure uniform distribution. The impregnated material was subjected to ultrasonication for 15 min to further enhance dispersion, dried at 110 °C for 4 h, and subsequently calcined at 500 °C for 4 h in air. After cooling to room temperature, the second impregnation was carried out using an aqueous solution of nickel nitrate hexahydrate, calculated to yield a 2.0 wt% NiO loading, following the same procedure (ultrasonication, drying, and calcination). The final catalyst precursors were denoted as NiMo/YX, for example, NiMo/YH-1. Prior to catalytic testing, these oxide precursors were sulfided in situ.

3.4. Characterization

XRD patterns were recorded on a Bruker D8 Advance diffractometer (Bruker AXS GmbH, Karlsruhe, Germany). N_2 physisorption was performed on a Micromeritics TriStar II 2020 analyzer (Micromeritics Instrument Corporation, Norcross, GA, USA). TEM was conducted on a Tecnai G2 F20 instrument (Thermo Fisher Scientific, Waltham, MA, USA). NH_3-TPD and H_2-TPR were carried out on a Tianjin Xianqua TP-5076 system (Tianjin Xianquan Instrument Co., Ltd., Tianjin, China). Py-FTIR spectra were recorded on a Digilab spectrometer (Digilab LLC, Cambridge, MA, USA). XPS analysis was performed on a Thermo Scientific K-Alpha+ spectrometer (Thermo Fisher Scientific, Waltham, MA, USA).

3.5. Catalytic Evaluation

Naphthalene hydrocracking was performed in a fixed-bed reactor. The catalyst was sulfided in situ using a solution of 3 wt% CS_2 in cyclohexane at 350 °C and 4.0 MPa for 4 h under a H_2 flow. The sulfidation reaction was carried out at a WHSV of 4 h^{-1}. The catalyst (0.5 g) was positioned in the middle of the reactor tube, sandwiched between layers of quartz sand with a catalyst-to-quartz sand mass ratio of 1:2.6. Reactions were conducted at 350–425 °C, 4.0 MPa, a WHSV of 4 h^{-1}, and an H_2/oil ratio of 600. The feedstock was 5 wt% naphthalene in n-octane. Liquid products were analyzed by GC-MS. Naphthalene conversion (X_{NAP}%) and the yield of BTX (Y_{BTX}%) were calculated as follows:

$$X_{NAP}\ (\%) = [1 - (\text{Moles of Naphthalene in Product/Moles of Naphthalene in Feed})] \times 100\%$$

$$Y_{BTX}\ (\%) = X_{NAP}\ (\%) \times (\text{Total Molar Selectivity of Benzene, Toluene, and Xylenes})/100$$

4. Conclusions

This study demonstrates that the engineering of hierarchical porosity in USY zeolites via postsynthetic strategies is a critical determinant of their ultimate performance in selective hydrocracking. The method of constructing the hierarchical structure profoundly influences both the acidity and metal function. Hydrochloric acid treatment engineered a hierarchical gradient pore acidity system that enforced a sequential "hydrogenation-then-cracking" pathway, yielding the highest BTX selectivity. Urea treatment enhanced acidity uniformly within a mildly desilicated hierarchical framework, leading to high activity

but slightly lower selectivity due to parallel reactions. The NaOH-surfactant method created a hierarchical network with ordered mesopores but simultaneously destroyed the strong acidity necessary for cracking and impaired metal sulfidation, resulting in poor performance. The findings highlight that hierarchical pore architecture design cannot be considered in isolation; it is intrinsically linked to the chemical functionality of the catalyst. For selective PAH hydrocracking, creating a spatially differentiated hierarchical environment that guides the reaction sequence is more effective than simply maximizing porosity or acidity uniformly.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/catal16010031/s1, Figure S1: A typical total ions chromatograph of naphthalene HCK liquid products on the NiMo/YH-1 catalyst. Reaction conditions: T = 425 °C, P = 4.0 MPa, H_2/Oil = 600 (volume ratio) and WHSV = 4 h^{-1}; Figure S2: TG curve of NiMo/YH-1-spent and NiMo/YN-spent; Table S1: The catalytic performance of series catalysts reported in the literatures.

Author Contributions: Conceptualization, X.C. and X.W.; Methodology, H.L.; Validation, M.L.; Investigation, M.L., P.N., J.M., Y.Z. and K.Z.; Data curation, H.L. and Y.Z.; Writing—original draft, H.L.; Writing—review and editing, P.N., J.M., X.C.; Supervision, X.W.; Funding acquisition, X.W. All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement: The original contributions presented in this study are included in the article/Supplementary Materials.

Conflicts of Interest: Author Jun Ma was employed by the Petrochina Lanzhou Petrochemical Company. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

References

1. Peng, C.; Liu, B.; Feng, X.; Du, Y.; Fang, X. Engineering dual bed hydrocracking catalyst towards enhanced high-octane gasoline generation from light cycle oil. *Chem. Eng. J.* **2020**, *389*, 123461. [CrossRef]
2. Zhang, M.; Song, Q.; He, Z.; Wang, Q.; Wang, L.; Zhang, X.; Li, G. Tuning the mesopore-acid-metal balance in Pd/HY for efficient deep hydrogenation saturation of naphthalene. *Int. J. Hydrogen Energy* **2022**, *47*, 20881–20893. [CrossRef]
3. Gordeeva, N.A.; Shesterkina, A.A.; Vikanova, K.V.; Kustov, A.L. Naphthalene and its derivatives hydrogenation for hydrogen storage: Comparative analysis of the role of noble and non-noble metal catalysts–a review. *Int. J. Hydrogen Energy* **2024**, *69*, 113–121. [CrossRef]
4. Gutiérrez, A.; Arandes, J.M.; Castaño, P.; Olazar, M.; Bilbao, J. Preliminary studies on fuel production through LCO hydrocracking on noble-metal supported catalysts. *Fuel* **2012**, *94*, 504–515. [CrossRef]
5. Upare, D.P.; Park, S.; Kim, M.; Jeon, Y.-P.; Kim, J.; Lee, D.; Lee, J.; Chang, H.; Choi, S.; Choi, W. Selective hydrocracking of pyrolysis fuel oil into benzene, toluene and xylene over CoMo/beta zeolite catalyst. *J. Ind. Eng. Chem.* **2017**, *46*, 356–363. [CrossRef]
6. Shin, J.; Oh, Y.; Choi, Y.; Lee, J.; Lee, J.K. Design of selective hydrocracking catalysts for BTX production from diesel-boiling-range polycyclic aromatic hydrocarbons. *Appl. Catal. A Gen.* **2017**, *547*, 12–21. [CrossRef]
7. Zhu, L.; Shan, S.; Petkov, V.; Hu, W.; Kroner, A.; Zheng, J.; Yu, C.; Zhang, N.; Li, Y.; Luque, R. Ruthenium–nickel–nickel hydroxide nanoparticles for room temperature catalytic hydrogenation. *J. Mater. Chem. A* **2017**, *5*, 7869–7875. [CrossRef]
8. Lee, J.; Choi, Y.; Shin, J.; Lee, J.K. Selective hydrocracking of tetralin for light aromatic hydrocarbons. *Catal. Today* **2016**, *265*, 144–153. [CrossRef]
9. Kuchinskaya, T.; Kniazeva, M.; Samoilov, V.; Maximov, A. In situ generated nanosized sulfide Ni-W catalysts based on zeolite for the hydrocracking of the pyrolysis fuel oil into the BTX fraction. *Catalysts* **2020**, *10*, 1152. [CrossRef]
10. Cao, Z.; Zhang, X.; Xu, C.; Huang, X.; Wu, Z.; Peng, C.; Duan, A. Selective hydrocracking of light cycle oil into high-octane gasoline over bi-functional catalysts. *J. Energy Chem.* **2021**, *52*, 41–50. [CrossRef]

11. Golubev, I.; Dik, P.; Kazakov, M.; Pereyma, V.Y.; Klimov, O.; Smirnova, M.Y.; Prosvirin, I.; Gerasimov, E.Y.; Kondrashev, D.; Golovachev, V. The effect of Si/Al ratio of zeolite Y in NiW catalyst for second stage hydrocracking. *Catal. Today* **2021**, *378*, 65–74. [CrossRef]
12. Anilkumar, M.; Loke, N.; Patil, V.; Panday, R. Hydrocracking of hydrotreated light cycle oil to mono aromatics over non-noble bi-functional (ni-w supported) zeolite catalysts. *Catal. Today* **2020**, *358*, 221–227. [CrossRef]
13. Oh, Y.; Noh, H.; Park, H.; Han, H.; Nguyen, T.-B.; Lee, J.K. Molecular-size selective hydroconversion of FCC light cycle oil into petrochemical light aromatic hydrocarbons. *Catal. Today* **2020**, *352*, 329–336. [CrossRef]
14. Jia, H.; Yan, W.; Dong, Q.; Qing, B.; Hao, W.; Du, Y.; Zhang, X.; Ma, J.; Li, R. Selective ring-opening in 9, 10-dihydrophenanthrene hydrocracking over a practical NiMo/Al_2O_3-USY catalyst under mild conditions. *J. Energy Inst.* **2024**, *116*, 101716. [CrossRef]
15. Zheng, K.; Liu, M.; Li, H.; Chen, X.; Wang, X. Optimizing Metal Sites in Hierarchical USY for Selective Hydrocracking of Naphthalene to BTX. *Molecules* **2025**, *30*, 4023. [CrossRef]
16. Peng, C.; Zhou, Z.; Cheng, Z.; Fang, X. Upgrading of light cycle oil to high-octane gasoline through selective hydrocracking over non-noble metal bifunctional catalysts. *Energy Fuels* **2019**, *33*, 1090–1097. [CrossRef]
17. Hu, Z.; Hu, P.; Wang, X.; Wu, T.; Ge, S.; Xie, H.; Liu, B.; Chen, S.; Wu, Z. Selective hydrocracking of 1-methylnaphthalene to benzene/toluene/xylenes (BTX) over NiW/Beta bifunctional catalyst: Effects of metal–acid balance. *Fuel* **2024**, *363*, 130947. [CrossRef]
18. Jurković, D.L.; Kostyniuk, A.; Likozar, B. Mechanisms, reaction micro-kinetics and modelling of hydrocracking of aromatic biomass tar model compounds into benzene, toluene and xylenes (BTX) over H-ZSM-5 catalyst. *Chem. Eng. J.* **2022**, *445*, 136898. [CrossRef]
19. Kong, X.; Li, Y.; Liu, Z.; Zou, Y.; Li, D.; Liu, J.; Wang, W.; Wang, C.; Xu, C.; Wang, X. Pore-Structure Engineering of Hierarchical Ni/Y Catalysts for the Enhanced Selective Hydrocracking of Naphthalene to BTX. *ACS Sustain. Chem. Eng.* **2025**, *13*, 8817–8828. [CrossRef]
20. Serrano, D.P.; Escola, J.; Pizarro, P. Synthesis strategies in the search for hierarchical zeolites. *Chem. Soc. Rev.* **2013**, *42*, 4004–4035. [CrossRef]
21. Serrano, D.; Escola, J.; Sanz, R.; Garcia, R.; Peral, A.; Moreno, I.; Linares, M. Hierarchical ZSM-5 zeolite with uniform mesopores and improved catalytic properties. *New J. Chem.* **2016**, *40*, 4206–4216. [CrossRef]
22. Linares, N.; Cirujano, F.G.; De Vos, D.E.; García-Martínez, J. Surfactant-templated zeolites for the production of active pharmaceutical intermediates. *Chem. Commun.* **2019**, *55*, 12869–12872. [CrossRef] [PubMed]
23. Sato, K.; Nishimura, Y.; Matsubayashi, N.; Imamura, M.; Shimada, H. Structural changes of Y zeolites during ion exchange treatment: Effects of Si/Al ratio of the starting NaY. *Microporous Mesoporous Mater.* **2003**, *59*, 133–146. [CrossRef]
24. Meng, F.; Ding, Y.; Meng, W.; Mi, G.; Qiu, J. Modification of molecular sieves USY and their application in the alkylation reaction of benzene with cyclohexene. *ChemistrySelect* **2020**, *5*, 8935–8941. [CrossRef]
25. Yu, H.; Guo, D.; Hao, W.; Li, Z.; Jiang, W.; Li, L.; Wang, X.; Subhan, F.; Song, Z.; Liu, Y. A selective alkaline etching strategy to prepare hierarchical Beta molecular sieves for the hydrodeoxygenation of phenol. *Fuel* **2023**, *334*, 126577. [CrossRef]
26. Wang, Q.; Hou, Z.-G.; Zhang, B.; Liu, J.; Song, W.-Y.; Xue, D.-S.; Liu, L.-Z.; Wang, D.; Chen, X.-G. Preparation of a highly efficient Pt/USY catalyst for hydrogenation and selective ring-opening reaction of tetralin. *Pet. Sci.* **2018**, *15*, 605–612. [CrossRef]
27. Chawla, A.; Linares, N.; Rimer, J.D.; García-Martínez, J. Time-resolved dynamics of intracrystalline mesoporosity generation in USY zeolite. *Chem. Mater.* **2019**, *31*, 5005–5013. [CrossRef]
28. Kong, X.; Liu, Z.; Zou, Y.; Wang, C.; Xu, C.; Wang, X. Preparation and optimization of dendritic Ni/Y-silica composite catalysts for naphthalene hydrocracking to BTX. *J. Catal.* **2025**, *450*, 116331. [CrossRef]
29. Fan, J.; Duan, A.; Xu, C.; Zhao, Z.; Zheng, P.; Wang, X.; Li, H.; Xiao, C.; Mei, J.; Jiang, G. Hydrodesulfurization Properties of Dibenzothiophene over NiMo Catalysts Supported on Cubic Fm 3 m Mesoporous Structure and High-Framework Aluminum-Modified AlKIT-5. *Energy Fuels* **2018**, *32*, 9793–9803. [CrossRef]
30. Kubička, D.; Kumar, N.; Venäläinen, T.; Karhu, H.; Kubičková, I.; Österholm, H.; Murzin, D.Y. Metal—support interactions in zeolite-supported noble metals: Influence of metal crystallites on the support acidity. *J. Phys. Chem. B* **2006**, *110*, 4937–4946. [CrossRef] [PubMed]
31. Lee, S.-U.; Lee, Y.-J.; Kim, J.-R.; Kim, E.-S.; Kim, T.-W.; Kim, H.J.; Kim, C.-U.; Jeong, S.-Y. Selective ring opening of phenanthrene over NiW-supported mesoporous HY zeolite catalyst depending on their mesoporosity. *Mater. Res. Bull.* **2017**, *96*, 149–154. [CrossRef]
32. Zhengkai, C.; Xia, Z.; Chunming, X.; Aijun, D.; Rong, G.; Zhen, Z.; Ziming, W.; Chong, P.; Jianmei, L.; Xilong, W. The Synthesis of Al-SBA-16 Materials with a Novel Method and Their Catalytic Application on Hydrogenation for FCC Diesel. *Energy Fuels* **2017**, *31*, 805–814.
33. Pedroarena, I.; Grande, L.; Torrez-Herera, J.J.; Korili, S.A.; Gil, A. Analysis by temperature-programmed reduction of the catalytic system Ni-Mo-Pd/Al_2O_3. *Fuel* **2023**, *334*, 126789. [CrossRef]

34. Zheng, L.; Li, Z.; Fu, P.; Sun, F.; Liu, M.; Guo, T.; Fan, Q. Development of Mo-modified pseudoboehmite supported Ni catalysts for efficient hydrogen production from formic acid. *ACS Omega* **2022**, *7*, 27172–27184. [CrossRef] [PubMed]
35. Nguyen, H.M.; Pham, G.H.; Ran, R.; Vagnoni, R.; Pareek, V.; Liu, S. Dry reforming of methane over Co–Mo/Al_2O_3 catalyst under low microwave power irradiation. *Catal. Sci. Technol.* **2018**, *8*, 5315–5324. [CrossRef]
36. Jiao, J.; Fu, J.; Wei, Y.; Zhao, Z.; Duan, A.; Xu, C.; Li, J.; Song, H.; Zheng, P.; Wang, X. Al-modified dendritic mesoporous silica nanospheres-supported NiMo catalysts for the hydrodesulfurization of dibenzothiophene: Efficient accessibility of active sites and suitable metal–support interaction. *J. Catal.* **2017**, *356*, 269–282. [CrossRef]
37. Chen, X.; Bartlam, C.; Lloret, V.; Moses Badlyan, N.; Wolff, S.; Gillen, R.; Stimpel-Lindner, T.; Maultzsch, J.; Duesberg, G.S.; Knirsch, K.C. Covalent Bisfunctionalization of Two-Dimensional Molybdenum Disulfide. *Angew. Chem. Int. Ed.* **2021**, *60*, 13484–13492. [CrossRef]
38. Wang, X.; Zhao, Z.; Zheng, P.; Chen, Z.; Duan, A.; Xu, C.; Jiao, J.; Zhang, H.; Cao, Z.; Ge, B. Synthesis of NiMo catalysts supported on mesoporous Al_2O_3 with different crystal forms and superior catalytic performance for the hydrodesulfurization of dibenzothiophene and 4,6-dimethyldibenzothiophene. *J. Catal.* **2016**, *344*, 680–691. [CrossRef]
39. Zhou, W.; Liu, M.; Zhou, Y.; Wei, Q.; Zhang, Q.; Ding, S.; Zhang, Y.; Yu, T.; You, Q. 4,6-Dimethyldibenzothiophene hydrodesulfurization on nickel-modified USY-supported NiMoS catalysts: Effects of modification method. *Energy Fuels* **2017**, *31*, 7445–7455. [CrossRef]
40. Chen, J.; Oliviero, L.; Portier, X.; Maugé, F. On the morphology of MoS_2 slabs on MoS_2/Al_2O_3 catalysts: The influence of Mo loading. *RSC Adv.* **2015**, *5*, 81038–81044. [CrossRef]
41. Kim, Y.-S.; Yun, G.-N.; Lee, Y.-K. Novel Ni_2P/zeolite catalysts for naphthalene hydrocracking to BTX. *Catal Commun.* **2014**, *45*, 133–138. [CrossRef]
42. Yu, F.; Zhang, C.; Geng, R.; Zhou, H.; Dong, Q.; Liu, S.; Fan, B.; Li, R. Hydrocracking of naphthalene over Beta zeolite coupled with NiMo/γ-Al_2O_3: Investigation of metal and acid balance based on the composition of industrial hydrocracking catalyst. *Fuel* **2023**, *344*, 128049. [CrossRef]
43. Dong, Q.; Zhang, C.; Zhang, H.; Yu, F.; Liu, S.; Fan, B.; Li, R. Design and preparation of Pt@ SSZ-13@ β core-shell catalyst for hydrocracking of naphthalene. *J. Catal.* **2023**, *421*, 365–375. [CrossRef]
44. Oh, J.; Choi, Y.; Shin, J.; Kim, K.; Lee, J.K. Synergistic shape selectivity of H-Beta and H-ZSM-5 for Xylene-rich BTX production by hydrocracking of heavy-aromatic compounds. *Fuel Process. Technol.* **2023**, *249*, 107856. [CrossRef]
45. Shen, Z.; He, P.; Wang, A.; Harrhy, J.; Meng, S.; Peng, H.; Song, H. Conversion of naphthalene as model compound of polyaromatics to mono-aromatic hydrocarbons under the mixed hydrogen and methane atmosphere. *Fuel* **2019**, *243*, 469–477. [CrossRef]

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